

Physical Characteristics of Bitumen in South Asia

COMPARATIVE TECHNICAL SPECIFICATIONS (BANGLADESH, INDIA, NEPAL, PAKISTAN)

Table 1: Standard Specification Format

PROPERTY	BANGLADESH (RHD / 60/70)	INDIA (MORT&H / 60/70)	NEPAL (DOR / 60/70)	PAKISTAN (NHA / 60/70)	TEST STANDARD
Penetration at 25°C, 100g, 5s	60 – 70 dmm	60 – 70 dmm	60 – 70 dmm	60 – 70 dmm	ASTM D5 / AASHTO T49
Softening Point	48 – 56 °C	47 – 55 °C	48 – 56 °C	48 – 56 °C	ASTM D36 / AASHTO T53
Ductility at 25°C	Min. 100 cm	Min. 75 cm	Min. 100 cm	Min. 100 cm	ASTM D113 / AASHTO T51
Flash Point	Min. 232 °C	Min. 232 °C	Min. 232 °C	Min. 232 °C	ASTM D92 / AASHTO T48
Specific Gravity at 25°C	1.01 – 1.06	1.01 – 1.06	1.01 – 1.06	1.01 – 1.06	ASTM D70 / AASHTO T228
Solubility in Trichloroethylene	Min. 99.0%	Min. 99.0%	Min. 99.0%	Min. 99.0%	ASTM D2042 / AASHTO T44

* Note: Values follow national highway authority standards for penetration grade bitumen.

Table 2: Alternative Regional Format (South Asian Comparison)

PROPERTY	BANGLADESH	INDIA	NEPAL	PAKISTAN
Bitumen Code	BD-Bit 1	IN-Bit 2	NP-Bit 3	PK-Bit 4
Bitumen Grade	60/70	VG-30 (50/70)	80/100	60/70
Penetration @ 25°C, 100g/5s	60 – 70 (0.1 mm)	50 – 70 (0.1 mm)	80 – 100 (0.1 mm)	60 – 70 (0.1 mm)
Softening Point	45 – 52 °C	Min 47 °C	42 – 51 °C	48 – 54 °C
Ductility @ 25°C	Min 100 cm	Min 40 cm	Min 100 cm	Min 100 cm
Absolute Viscosity @ 60°C	—	2400 – 3600 Poise	—	—
Flash Point	Min 250 °C	Min 222 °C	Min 225 °C	Min 232 °C

* Note: Derived from regional comparative physical properties sample breakdown.